

ESTIMATION OF 2-D DOA JOINT FREQUENCY OF SIGNAL VIA A SINGLE VECTOR HYDROPHONE

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Abstract: Based on a single vector hydrophone, two sub-matrix of the same formation which formed signal data matrix was constructed by delayed time data. On the basis of LS-ESPRIT, the signal data matrix was decomposed by one eigenvalue decomposition to get its eigenvalues and corresponding eigenvectors, which included the information of frequency and angle, so two-dimensional DOA and frequency joint estimation was realized for narrow-band signal. This method had better estimation accuracy, which didn't need any search procedure, the three-dimensional parameter matching was automatically achieved by this method. Compare with DOA matrix method via a single vector hydrophone and LS-ESPRIT method via acoustic pressure array by MATLAB, the results showed that this method have better estimation performance.

Keywords: A single vector hydrophone; LS-ESPRIT; 2-D DOA estimation; Frequency estimation

1 Introduction

In recent decades, the estimation of multi-dimensional parameters of signal was more and more adverted, and it became a very active area of research in the field of signal processing. Compared to the estimation of signal multi-dimensional parameters, people have acquire more significant achievement on the estimation of one-dimensional DOA, but if we want get the DOA of signal, we must know the frequency of signal, for unknown environment of signal, the research of estimating frequency and DOA is always combined. The vector hydrophone has a unique advantage on the estimation of signal parameters, which can measure the information of pressure and velocity in the sound field. In Ref. [1], FENG advanced to use acoustic energy flow to estimation DOA, which can only estimate a source of orientation. HE use a single vector hydrophone to realize joint estimation of 2-D DOA and frequency in Ref. [2], which based on the DOA matrix method, but it did not detailed discuss the selection of the delay and the estimated performance of the algorithm needs to be improved. JIA use the antenna array to estimate 2-D DOA and frequency [3], the method can be used on the scalar acoustic array, but the method parameter matching process is relatively complicated and it need large-scale array. This paper presented a new method to realize the joint estimation of signal 2-D DOA and frequency by a single vector hydrophone and made a

brief analysis focusing on the delay for the frequency estimation. Two formations of the same sub-array were constructed by the delay gene, The LS-ESPRIT [4-5] method was used to actualize the joint estimation of 2-D DOA and frequency of single-frequency signal. The simulation results show that relative to the DOA matrix method, the performance of the proposed method is much better, while compare to LS-ESPRIT method based on the scalar acoustic array, the lineup size of a single vector hydrophone is much smaller.

The rest of this paper is organized as follows: section 2 introduces the signal model and the proposed algorithm. Section 3 presents the acoustic pressure array with LS-ESPRIT algorithm and three-dimensional parameter matching process. In section 4, numerical simulations demonstrating the better parameter estimation accuracy of the proposed algorithm than the others are presented. Section 5 gives the final conclusions of this paper.

2 Signal model and algorithm

Assume that the number of single frequency signal is K and the center frequency is f_k ($i = 1, \dots, K$), Assume that the 2-D DOA of each plane wave in the isotropic homogeneous fluid medium are $\{\theta_k, \varphi_k, (k = 1, \dots, K)\}$, $\theta_k \in [0, 2\pi]$ is the azimuth of the k th signal and $\varphi_k \in [0, \pi]$ is the elevation of the k th signal. The relation between signal and noise is that the signals are not correlation and the signals and the noises are not correlation too, besides the noises are statistically independent complex Gaussian white noise with zero mean.

The specific form of signal is

$$p_k(t) = \sqrt{E_k} \exp(j(2\pi f_k t + \beta_k)) \quad (1)$$

Where E_k is Signal power, f_k is signal frequency, β_k is random delay, which belongs to $[0, 2\pi]$ and is obedience to uniform distribution.

The output of a single vector hydrophone at time t is expressed as follows:

$$Y_1(t) = A_1 P(t) + N_1(t) \quad (2)$$

Add the delay τ ($\tau = n\tau_s$, $n \geq 1$) on t , we can obtain:

$$Y_2(t) = A_2 P(t) + N_2(t) \quad (3)$$

From Eqs. (2) and (3), we have

$$Y(t) = \begin{bmatrix} Y_1(t) \\ Y_2(t) \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} P(t) + \begin{bmatrix} N_1(t) \\ N_2(t) \end{bmatrix} = A P(t) + N(t) \quad (4)$$

$$P(t) = [p_1(t) \cdots p_K(t)]^T$$

$$A_1 = [a_1 \ a_2 \cdots a_K]$$

$$A_2 = [a_1 \exp(j2\pi f_1 \tau) \cdots a_K \exp(j2\pi f_K \tau)] = A_1 \Omega$$

$$a_k = [1 \ u_k \ v_k \ w_k]^T$$

$$\Omega = \text{diag}(\exp(j2\pi f_1 \tau) \cdots \exp(j2\pi f_K \tau))$$

Where T denotes the transpose, a_k is the steering vector, $N_1(t)$ and $N_2(t)$ are the different times noise received by a single vector hydrophone. The diagonal element of Ω include the signals' frequencies, which are called the frequency factor.

Define the autocorrelation matrix of $Y(t)$ as follows:

$$R_s = E[Y(t)Y(t)^H] \quad (5)$$

we can obtain eight eigenvalues which are $\lambda_1 \geq \lambda_2 \geq \cdots \lambda_k \geq \lambda_{k+1} = \cdots = \lambda_8 = \sigma_n^2$ from R_s by eigenvalue decomposition, the corresponding eigenvalue vectors are $e_1, e_2, \cdots, e_k, e_{k+1}, \cdots, e_8$, $e_1, e_2, \cdots, e_k$ spanned the subspace denote the signal subspace as

$$E = [e_1, e_2, \cdots, e_k] = A T \quad (6)$$

Where T is a non-singular matrix. From (10), we can have

$$E = \begin{bmatrix} E_1 \\ E_2 \end{bmatrix} \quad (7)$$

$$E_1 = A_1 T, \quad E_2 = A_2 T = A_1 \Omega \quad (8)$$

There must be a similarity matrix Ψ for the diagonal matrix Ω , so we get

$$\Psi = T^{-1} \Omega T \quad (9)$$

From Eqs. (13) and (14), we can obtain a expression, that is, $E_1 \Psi = E_2$. If there is no noise, E_1, E_2 are replaced by $\hat{E}_1, \hat{E}_2$. LS-ESPRIT algorithm is used on Eq. (9), that is,

$$\hat{\Psi}_{LS} = (\hat{E}_1^H \hat{E}_2)^{-1} \hat{E}_1^H \hat{E}_2 \quad (10)$$

From Eq. (13), $\hat{\Omega}$ and $\hat{T}$ are obtained by decomposing $\hat{\Psi}_{LS}$ and $\hat{A}_1$ is expressed as follows:

$$\hat{A}_1 = [\hat{a}_1 \ \hat{a}_2 \cdots \hat{a}_K] = \frac{1}{2} \{ \hat{E}_1 \hat{T}^{-1} + \hat{E}_2 \hat{T}^{-1} \hat{\Omega}^{-1} \} \quad (11)$$

Accordingly, direction cosines are expressed as

$$\hat{u}_k = \frac{[\hat{a}_K]_1}{\sqrt{[\hat{a}_K]_1^2 + [\hat{a}_K]_2^2 + [\hat{a}_K]_3^2}} \quad (12)$$

$$\hat{v}_k = \frac{[\hat{a}_K]_2}{\sqrt{[\hat{a}_K]_1^2 + [\hat{a}_K]_2^2 + [\hat{a}_K]_3^2}} \quad (13)$$

$$\hat{w}_k = \frac{[\hat{a}_K]_3}{\sqrt{[\hat{a}_K]_1^2 + [\hat{a}_K]_2^2 + [\hat{a}_K]_3^2}} \quad (14)$$

From (18)~(20), the 2-D DOA and frequency of signals are estimated, that is,

$$\hat{\theta}_k = \angle(\hat{u}_k + j \hat{v}_k) \quad (15)$$

$$\hat{\varphi}_k = \arcsin \left(\sqrt{\hat{u}_k^2 + \hat{v}_k^2} \right) = \arccos(\hat{w}_k) \quad (16)$$

$$\hat{f}_k = \frac{\arg[\hat{\Omega}(k, k)]}{2\pi\tau} \quad (17)$$

$k=1, 2, \cdots, K$. It is explained that $|f_p - f_q| \neq \frac{l}{\tau}$ (l is an arbitrary integer, $p, q \in (1 \cdots K)$ 且 $p \neq q$), but we note that τ must be satisfied the condition $\tau \leq \frac{1}{2f_m}$

(f_m is the maximum one of frequencies of signals), or the estimation of frequency will be failed. The reason is that the frequency of signals is estimated by the phase.

3 Acoustic pressure array structure and Algorithm

In Figure 1, we assume that c represent the speed of sound in the water, r_k represents the delay factor of the k th signal and g_k and h_k represent respectively the space phase shift factor at axis X and axis Y.

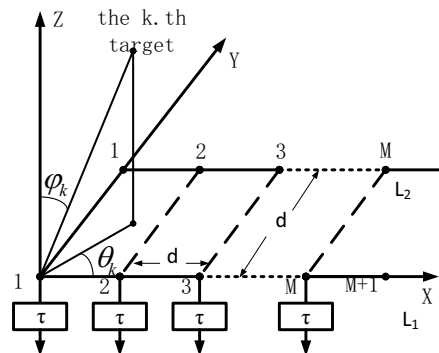


Figure 1 The structure of acoustic pressure array

$$r_k = \exp(j2\pi f_k \tau)$$

$$g_k = \exp(j2\pi f_k d \cos \theta_k \sin \varphi_k / c)$$

$$h_k = \exp(j2\pi f_k d \sin \theta_k \sin \varphi_k / c)$$

$$\begin{aligned}
 X_0(t) &= B_x P(t) \\
 X_1(t) &= B_x \Phi_x P(t) \\
 X_2(t) &= B_x \Phi_y P(t) \\
 X_3(t) &= B_x \Phi_f P(t) \\
 B_x &= [b_1 b_2 \cdots b_K] \\
 b_k &= [1 g_k \cdots g_k^{M-1}]^T \\
 \Phi_x &= \text{diag}(g_1 g_2 \cdots g_K) \\
 \Phi_y &= \text{diag}(h_1 h_2 \cdots h_K) \\
 \Phi_f &= \text{diag}(r_1 r_2 \cdots r_K) \\
 k &= 1, 2, \dots, K
 \end{aligned}$$

Data matrix is constructed as follows:

$$X(t) = \begin{bmatrix} X_0(t) \\ X_1(t) \\ X_2(t) \\ X_3(t) \end{bmatrix} \quad (18)$$

The corresponding cross-correlation matrix is

$$R = E[X(t)X(t)^H] \quad (19)$$

$\hat{\Phi}_x$ can be obtained by decomposing R [6-8], from which we can obtain the 2-D DOA and frequency, that is,

$$\hat{f}_k = \left| \frac{\arg(\hat{r}_k)}{2\pi\tau} \right| \quad (20)$$

$$\hat{\theta}_k = \arctan \left(\frac{\arg(\hat{h}_k)}{\arg(\hat{g}_k)} \right) \quad (21)$$

$$\hat{\varphi}_k = \arccos \left(\frac{c}{2\pi \hat{f}_k d} \sqrt{(\arg(\hat{g}_k))^2 + (\arg(\hat{h}_k))^2} \right) \quad (22)$$

$k = 1, 2, \dots, K$.

4 Simulation and analysis

Assume that $p_1(t)$ ($f_1 = 150\text{Hz}$, $E_1 = 1$) and $p_2(t)$ ($f_2 = 100\text{Hz}$, $E_2 = 1$) are the two independent signals, the azimuth and elevation angles of incidence of the two signals are separately $(50^\circ, 80^\circ)$ and $(30^\circ, 50^\circ)$. $f_s = 1000\text{Hz}$ and $\tau = 0.003\text{s}$, the numbers of the acoustic pressure array are respectively 5, 7 and 9.

First experiment: assume that the snapshot $N = 1000$, the value range of SNR are from 0dB to 30dB, 200 Monte Carlo simulation calculation are executed every time. The results are showed in Figure 2, Figure 3 and Figure 4.

Second experiment: assume that $\text{SNR} = 20\text{dB}$ and the value range of N are from 100 to 1000, 200 Monte Carlo simulation calculations are executed every time. The results are showed in Figure 5, Figure 6 and Figure 7.

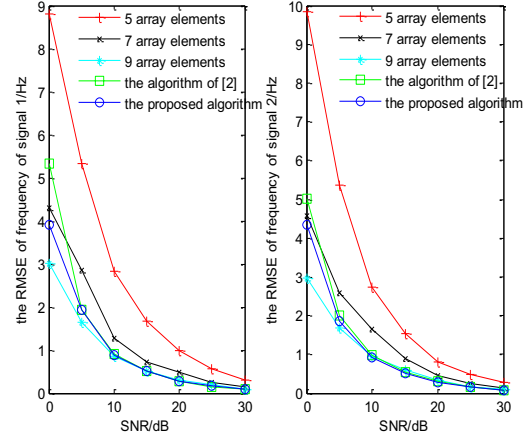


Figure 2 The RMSE of frequency via SNR

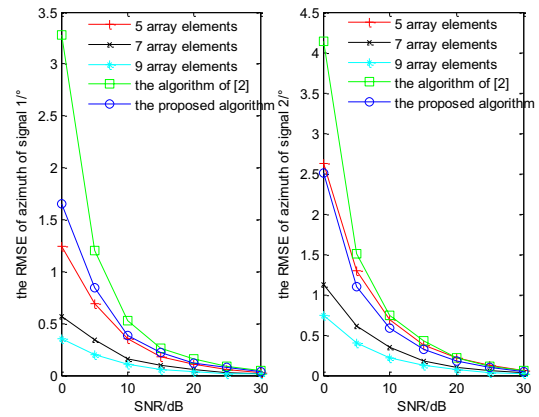


Figure 3 The RMSE of azimuth via SNR

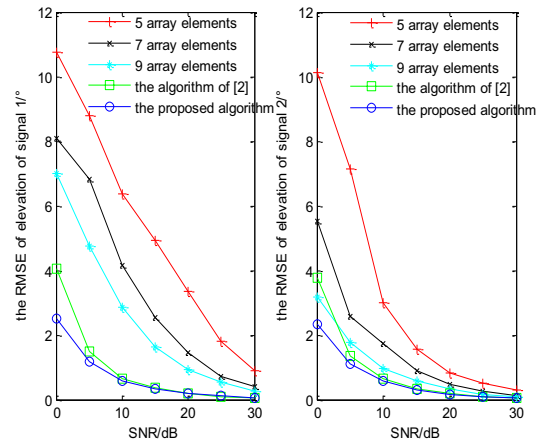


Figure 4 The RMSE of elevation via SNR

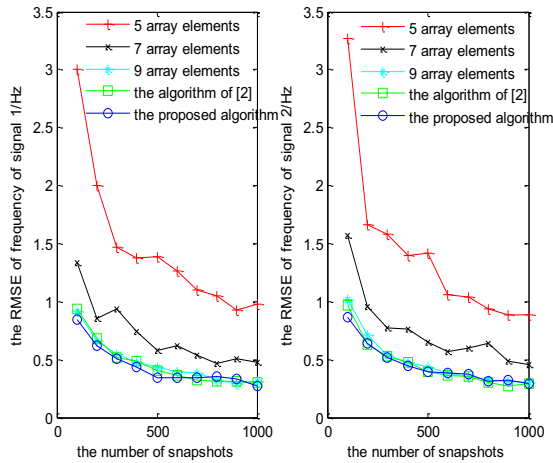


Figure 5 The RMSE of frequency via snapshots

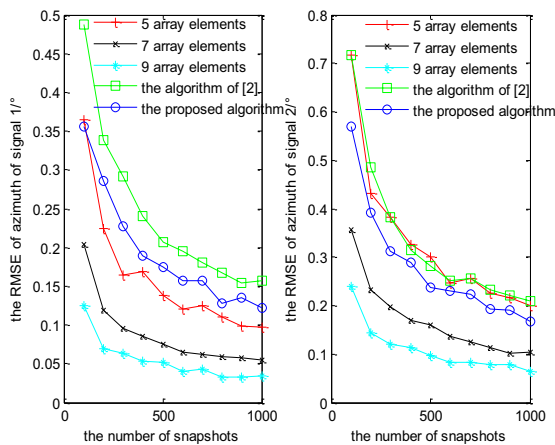


Figure 6 The RMSE of azimuth via snapshots

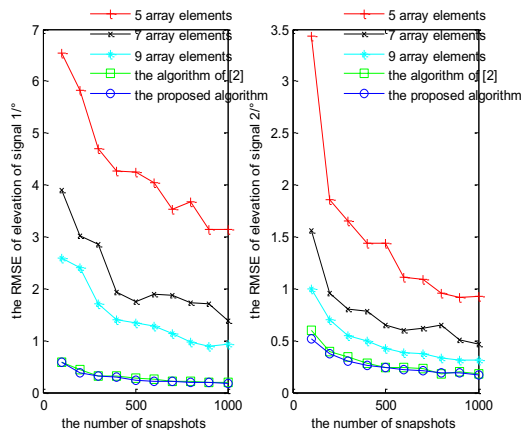


Figure 7 The RMSE of elevation via snapshots

Two simulation results show that the proposed algorithm exhibit good estimation accuracy of signal parameter at low SNR, the estimation accuracy gradually gained by increasing the signal to noise ratio and the number of the snapshots.

From the two simulations, we could obtain the outcome as follows:

1) It can be seen from Figures 2 and 5 that the frequency estimation accuracy of the proposed algorithm is higher than the algorithm of Ref. [2] in the case of low SNR, and its overall estimated performance hold the line with acoustic pressure array owned nine array elements.

2) From Figures 3 and 6, it is clear that the azimuth estimation accuracy of proposed algorithm is higher than the Ref. [2] algorithm, because the size of delay factor is determined by the high-est frequency of the signals, so the estimated accuracy of the azimuth of the signal 1 is higher than Signal 2.

3) In Figures 4 and 7, it shows that the elevation estimation accuracy of proposed algorithm is higher than the algorithm of Ref. [2], and significantly higher than the algorithm of acoustic pressure array. From Eq. (22), it is clear that the estimation accuracy of elevation is enslaved to the estimation accuracy of frequency, which led to the estimation accuracy of the signal elevation seriously decline. However, the elevation estimation accuracy of the proposed algorithm is unacted on the other parameters estimation accuracy.

5 Conclusions

The signal parameter estimation of the proposed algorithm is implemented by transforming the received signal data with a certain matrix transformation. It is proved by experiments that the parameter estimation accuracy of the proposed algorithm is higher than the algorithm of literature [2]. Moreover, compare to acoustic pressure array with nine array element, the complexity of the proposed algorithm is more smaller, and its performance of the azimuth and frequency estimation is balance to the acoustic pressure array with nine array element and the proposed algorithm provided more higher elevation estimation accuracy. Besides, a single vector hydrophone has a distinct advantage in size, especially in the low frequency.

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